



Erwan Hochart

PhD Candidate

Inquisitive, hard-working and determined, I am currently looking for a role in an environment that will challenge and motivate me to continue developing my analytical, numerical and computational skills while letting my curiosity flourish.

- 27th June, 1998. Clamart, France.
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- LinkedIn Page
- GitHub Page
- Personal Site

Languages

- English Bilingual
- French Bilingual
- Chinese HSK3/4

Soft Skills

- Analytical
- Attention to detail
- Communication
- Curious
- Organisation
- Problem-solving

Programming Skills

- Python Fluent
- C++ Proficient
- Bash Proficient
- Parallel Programming Competent
- HPC Competent
- Machine Learning Familiar

Education

- Sept. 2020 - Feb. 2023 **Master's of Physics** Leiden University, the Netherlands
Specialisation: Research in Physics, Cosmology
- Sept. 2017 - May 2020 **Bachelor's in Astronomy** Rijksuniversiteit Groningen, the Netherlands
Major: Astronomy
- Sept. 2016 - May 2017 **Foundation Year of Physics** University of Manchester, England
- Aug. 2002 - May 2016 **Secondary Education** Western Academy of Beijing, China
International Baccalaureate

Experience

- Sept. 2023 - **PhD: Computational Astronomy** Leiden University, the Netherlands
 - Programming languages: C++, Python, Bash, Slurm.
 - HPC and parallelisation including use of University and National supercomputers.
 - Presentation of results (poster, talks, scientific papers).
 - Teacher assistant for Master course 'Simulation and Modeling in Astrophysics'.
 - Grade Msc students' final thesis presentation.
 - Additional Courses: Machine Learning I and II, Software Engineering, Software Development.
- Nov. 2019 - Feb. 2020 **Teaching Assistant** Rijksuniversiteit Groningen, the Netherlands
 - Course: 'The Evolving Universe', for Bachelor students
 - Organise tutorial sessions.
 - Weekly presentations summarising previous class.

Publications: [See Google Scholar](#)

- May 2026 **Oort Cloud Ecology - IV. Exchanging Asteroids**
Astronomy & Astrophysics
[Link to the paper](#)
- April 2026 **Nemesis: A Multi-Scale, Multi-Physics Algorithm for Astrophysics**
Astronomy & Astrophysics
[Link to the paper](#) + [GitHub repository](#)
- July 2025 **Why Wide Jupiter-Mass Binary-Objects Cannot Form**
Nature Astronomy
[Link to the paper](#)
- Mar. 2024 **The origin of wide Jupiter Mass Binary Objects**
SciPost
[Link to the paper](#) + [GitHub repository](#)
- Feb. 2024 **The Steady State of Black Holes in Galactic Nuclei**
Astronomy & Astrophysics
[Link to the paper](#) + [GitHub repository](#)

Conference Contributions

May 2026	Exchanging Asteroids in Star Clusters <i>NAC 2026, Egmond aan Zee, the Netherlands</i> Link to abstract
September 2025	Nemesis: An Algorithm for Simulating Planetary Systems in Clusters <i>EPSC-DPS, Helsinki, Finland</i> Link to abstract
June 2025	Exchanging Asteroids in Star Clusters <i>Oort Cloud Meeting, Stockholm, Sweden</i> Link to poster
June 2024	The origin of wide Jupiter Mass Binary Objects <i>Exoplanets5, Leiden, the Netherlands</i> Link to poster
Oct. 2023	The Steady State of Black Holes in Galactic Nuclei <i>Belgian-Dutch GW Meeting, Maastricht, the Netherlands</i> Link to abstract + presentation

Projects

Sept. 2022 - Feb. 2023	Second Master's Thesis: Steady-State of Intermediate Mass Black Holes.	Leiden University
	• Computational research. • Python, C++, Slurm, Linux environment • Write an academic report. • Oral thesis defense. Link to the project.	
Aug. 2021 - Feb. 2022	First Master's Thesis: Testing Modified Gravity Through Gravitational Waves	Leiden University
	• Theoretical research. • Python, Linux environment • Write an academic report. • Oral thesis defense. Link to the project.	
June 2021 - Aug. 2021	Leiden/ESA Astrophysics Program	Leiden University, ESTEC and DLR
	• Literature analysis. • Write a report on the benefits of a Lunar radio telescope. Link to the report.	
Apr. 2020 - June 2020	Bachelor's Thesis: Cosmic Web Classification: NEXUS+ vs. DisPerSE	Rijksuniversiteit Groningen
	• Theoretical research. • Python, Slurm, Linux environment • Write an academic report. • Oral thesis defense. Link to the project.	

Supervision Experience

Feb. 2025 - Nov. 2025	Planet-moon ejections in close stellar encounters Msc student Yannick Badoux.	Leiden University, the Netherlands
Feb. 2025 - Nov. 2025	Slingshotting Black Holes Msc student Benson Rugers.	Leiden University, the Netherlands

Hobbies

Football, Reading, Scuba Diving, Travelling, Bird Watching, Photography, Gardening